FYUGP B.Sc. PHYSICS HONOURS

ELECTRODYNAMICS II

Semester V · Core in Major

MODULE 3: ELECTROMAGNETIC WAVES

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

Sinusoidal waves

1 Write the relation between the wavelength λ and the wave number k of a sinusoidal wave. **(1 mark)**

2 Write the relation between the speed v, the frequency ν and the wavelength λ of a wave. **(1 mark)**

Polarisation

3 State whether true or false: in a circularly polarised wave the two perpendicular components of $\vec{E}$ have equal amplitudes and differ in phase by $90°$. **(1 mark)**

4 Write the polarisation vector $\hat{n}$ of a wave polarised at angle θ to the x axis, in the xy plane. **(1 mark)**

Refractive index

5 Write the index of refraction n of a linear medium in terms of its permittivity ε and permeability μ. **(1 mark)**

Radiation pressure

6 Write the radiation pressure on a perfect absorber in terms of the intensity I and the speed of light c. **(1 mark)**

Section B: 2 marks each, answer all

Complex notation

7 Write the wave $f(z,t) = A\cos(kz - \omega t + \delta)$ in complex notation. State how the physical wave is recovered from the complex form. **(2 marks)**

Energy density of a plane wave

8 Write the energy density of a monochromatic plane wave of amplitude E_0 in vacuum, averaged over one cycle. **(2 marks)**

Waves in a linear medium

9 Write the speed of a monochromatic wave in a linear medium of index n, and the relation between the amplitudes B_0 and E_0 in that medium. **(2 marks)**

Normal incidence

10 Light passes normally from air ($n_1 = 1$) into glass ($n_2 = 1.5$). Find the fraction of the incident intensity that is reflected, and the fraction transmitted. **(2 marks)**

Brewster's angle

11 State Brewster's law for the angle θ_B at which the reflected wave is extinguished, for $\mu_1 \approx \mu_2$. Find θ_B for air to glass with $n_2 = 1.5$. **(2 marks)**

Section C: 4 marks each, answer any two

Waves in a linear medium

12 A monochromatic plane wave of frequency 5×10^{14} Hz travels in glass of index 1.5 (take $\mu = \mu_0$). Find its speed in glass, its wavelength in vacuum, and its wavelength in glass. **(4 marks)**

Radiation pressure

13 Sunlight of intensity 1300 W/m^2 falls normally on a perfect absorber, and then on a perfect reflector. Find the radiation pressure in each case. Take $c = 3.0 \times 10^8$ m/s. **(4 marks)**

Intensity and energy density

14 A monochromatic plane wave in vacuum has electric amplitude $E_0 = 100$ V/m. Find the magnetic amplitude B_0, the average intensity, and the average energy density. Take $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m and $c = 3.0 \times 10^8$ m/s. **(4 marks)**

Reflection and refraction

15 Light in air ($n_1 = 1.00$) strikes a glass surface ($n_2 = 1.50$) at an angle of incidence $40°$. Find the angle of reflection and the angle of refraction. **(4 marks)**

Section D: 10 marks each, answer any one

Electromagnetic waves in a linear medium

16 In a linear homogeneous medium with no free charge or current, derive the wave equation for $\vec{E}$ from Maxwell's equations, and show that $v = 1/\sqrt{\varepsilon\mu} = c/n$. For a plane wave, show that $B_0 = E_0/v$ and $I = \frac{1}{2} v \varepsilon E_0^2$. **(10 marks)**

Reflection and refraction, Brewster's angle

17 Light passes from medium 1 to medium 2. From the boundary conditions, state the laws of reflection and refraction, and derive Brewster's law $\tan\theta_B = n_2/n_1$ for $\mu_1 \approx \mu_2$. State the polarisation of the reflected beam at θ_B. **(10 marks)**

MODULE 3: MODEL ANSWERS

For checking your work. Each answer shows the key step and the result.

Section A. A1 $\lambda = 2\pi/k$. A2 $v = \nu\lambda$. A3 True: a circularly polarised wave has equal components, $90°$ apart in phase. A4 $\hat{n} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y}$ (Eq. 9.38). A5 $n = \sqrt{\varepsilon\mu/\varepsilon_0\mu_0} = c/v$. A6 $P = I/c$ (Eq. 9.65 for the absorber).

Section B. B7 $\tilde{f} = \tilde{A}e^{i(kz-\omega t)}$ with $\tilde{A} = Ae^{i\delta}$; the physical wave is $f = \mathrm{Re}(\tilde{f})$ (Eqs. 9.16–9.18). B8 $\langle u\rangle = \frac{1}{2}\varepsilon_0 E_0^2$, since the average of $\cos^2$ over a cycle is $\frac{1}{2}$ (Eq. 9.61). B9 $v = c/n$; $B_0 = E_0/v = nE_0/c$ (Eqs. 9.48, 9.69, 9.74). B10 $R = \left(\frac{n_1-n_2}{n_1+n_2}\right)^2 = 0.04$, and $T = 1 - R = 0.96$ (Eqs. 9.87, 9.88). B11 $\tan\theta_B = n_2/n_1 = 1.5$, so $\theta_B \approx 56.3°$ (Eq. 9.113).

Section C. C12 $v = c/n = 2.0 \times 10^8$ m/s; vacuum wavelength $\lambda_0 = c/\nu = 6.0 \times 10^{-7}$ m; wavelength in glass $\lambda = v/\nu = 4.0 \times 10^{-7}$ m. C13 Absorber: $P = I/c = 1300/3.0 \times 10^8 = 4.3 \times 10^{-6}$ Pa. Reflector: $P = 2I/c = 8.7 \times 10^{-6}$ Pa. C14 $B_0 = E_0/c = 3.3 \times 10^{-7}$ T; $I = \frac{1}{2}c\varepsilon_0 E_0^2 = 13.3$ W/m^2; $\langle u\rangle = I/c = 4.4 \times 10^{-8}$ J/m^3. C15 Reflection angle $= 40°$. Refraction: $\sin\theta_T = \sin 40°/1.50 = 0.429$, so $\theta_T \approx 25.4°$.

Section D. D16 Curl of Faraday's law and the Ampère–Maxwell law with $\nabla\cdot\vec{E} = 0$ give $\nabla^2\vec{E} = \mu\varepsilon\,\partial^2\vec{E}/\partial t^2$, so $v = 1/\sqrt{\varepsilon\mu} = c/n$ (Eqs. 9.68–9.69). For a plane wave, $\nabla\times\vec{E} = -\partial\vec{B}/\partial t$ gives $kE_0 = \omega B_0$, so $B_0 = E_0/v$. The average Poynting vector is $\langle S\rangle = E_0^2/(2\mu v) = \frac{1}{2}v\epsilon E_0^2$, using $1/\mu = v^2\varepsilon$ (Eqs. 9.72–9.74). D17 Matching the exponential factors at the boundary gives $\theta_I = \theta_R$ and $\sin\theta_I/\sin\theta_T = n_2/n_1$ (Eqs. 9.100–9.101). For polarisation in the plane of incidence the Fresnel reflection amplitude has $\alpha = \beta$ at θ_B, with $\sin^2\theta_B = (1-\beta^2)/((n_1/n_2)^2 - \beta^2)$; for $\mu_1 \approx \mu_2$, $\tan\theta_B = n_2/n_1$ (Eqs. 9.110–9.113). The reflected beam at θ_B is polarised perpendicular to the plane of incidence (footnote 19).

Mapping to the syllabus and textbook

Syllabus item	Topic	Griffiths section	Model questions
9	Waves in one dimension, sinusoidal waves, polarisation	9.1.1, 9.1.2, 9.1.4	1, 2, 3, 4, 7
10	Wave equation for $\vec{E}$ and $\vec{B}$	9.2.1	16
11	Monochromatic plane waves; $\vec{E}$ and $\vec{B}$ amplitudes	9.2.2	16, 9
12	Poynting's theorem and the Poynting vector	8.1.2	8, 14
13	Energy and momentum; radiation pressure	9.2.3	6, 8, 13, 14
14	Propagation in linear media; reflection, refraction and Brewster's angle (9.3.2 and 9.3.3 are background)	9.3.1, 9.3.2, 9.3.3	5, 9, 10, 11, 12, 15, 17